Skills needed

In terms of skill sets, since this is an interdisciplinary field, you need to be strong in the fundamentals from at least 3 or 4 fields, such as:

- Solid-state Physics
- Circuit Design and VLSI
- Computer Science and Neural Networks
- Computational Neurobiology

You need not be an expert in all these fields to get into neuromorphic computing. You need to be very focused on what information is required for your implementation, and hence good guidance is important in this area. I was fortunate enough to have been guided by eminent experts such as Prof. Dmitri Strukov, Prof. M. Jagadesh Kumar, and Prof. Manan Suri.

the fourth fundamental element apart from resistors, inductors, and capacitors. Many people gave some theoretical models for a memristor. My postdoc advisor Prof. Dmitri Strukov was among the people who invented memristors. He fabricated a memristor and proved that it shows all the properties (such as plasticity, STDP learning rule, etc) which could mimic the processing elements such as synapses and neurons in the human brain and could be harnessed for neuromorphic computing.

You worked on a neuromorphic chip at UCSB just two years ago. Could you elaborate on your experience there and what your team worked on?

The whole world is going through an artificial intelligence revolution. Neural networks are the driving force behind the artificial intelligence revolution. Neural networks exist in different flavours—artificial neural networks, SNNs, DNNs, CNNs, etc, which are efficient in performing different applications. However, the fundamental operation in all the neural networks is the vector-by-matrix multiplication or, in other words, multiply-accumulate operation. If we can somehow accelerate the vector-by-matrix multiplication, if we can execute this operation in an energy-efficient way, then we can realise ultra-low power hardware implementation of all neural networks. This is something that our group at UCSB realised and we developed different circuits and systems exploiting emerging non-volatile memories to perform vector-by-matrix multiplication in an efficient manner.

We utilised two kinds of non-volatile memories that were technologically advanced—one of them was flash memory. Flash memory is ubiquitous nowadays, ranging from memory cards and USB sticks to even cloud storage. In almost all devices in the internet of things (IoT) ecosystem, you'll find flash memory. They are already technologically mature. So, if these devices could be used for neuromorphic computing, if they could process information themselves, it would be one of the milestones in realising an ultra-energy-efficient green society. Currently, flash memories have the capability to store the data but there doesn't exist a conventional circuit or architecture with which you can make them process this data.

> THE MOST PROMISING CANDIDATE, CONSIDERING THE PRESENT TECHNICAL KNOWHOW, IS THE MEMRISTOR.

Another advantage of flash devices is that they are very dense. You can just imagine the density by looking at this scenario: In 2010, when I was purchasing my first phone, the internal memory was of the order of a few MB. And now, if you go for any smartphone, the internal memory (which is again a flash memory) is at least 64 or 128GB. The density has increased by more than a thousand times. So, if we can introduce processing power to flash, we can enable all smart devices to compute in-situ.

I also worked with oxide based resistive RAMs or memristors. We found out that they were even more promising with respect to energy efficiency. Probably, they will be able to achieve the levels of efficiency of the human brain for certain applications soon. We were collaborating with Applied Materials and Philips, who agreed to fabricate those chips for us.

You take a class on neuromorphic computing at IIT Kanpur. And you also have a research group called NeuroCHaSe. How has been the response of students in your class?

I introduced this course at IIT Kanpur immediately after I joined (one year ago). Although it was a PG-level course, I saw great participation from the UGs. In general, in PG courses, there are around 15-20 students. But in this course, there were 55 students, out of which around 20 students were from B.Tech. That was a very encouraging response and throughout the course they were very interactive. The huge B.Tech participation in my course bears testimony to the fact that undergrads are really interested in this topic. I believe that this could be introduced as one of the core courses in the UG curriculum in the future.

During the course, I introduced